

Department of Leather Engineering
Khulna University of Engineering & Technology
 B. Sc. Engineering 3rd Year 1st Term Examination 2023
LE 3107
Dyeing and Finishing I

Time: 3.0 Hours

Full Marks: 210

- N.B. i) Answer any **THREE** questions from each section in separate scripts.
 ii) Figures in the right margin indicate full marks.
 iii) Assume reasonable data if any missing.

SECTION-A

- 1(a) Which color will you observe if you see a Magenta-colored leather under yellow light? 08
 Explain.
- 1(b) Why auxochrome is required in a dye? Explain with proper example. 07
- 1(c) "Benzene is colorless, Nitro-benzene is pale yellow and p-nitro aniline is dark yellow", 12
 Explain the phenomenon according to Valence Bond Theory.
- 1(d) How absorption of radiation is responsible for color production? 08
- 2(a) Explain the effect of conjugation and resonance on color deepening of a compound. 10
- 2(b) Describe the different types of electronic transition from a bonding orbital to an anti- 10
 bonding orbital by a molecule with proper diagram.
- 2(c) Differentiate between phosphorescence and fluorescence by using Jablonski diagram. 10
- 2(d) Discuss the influencing factors of leather dyeing. 05
- 3(a) State the industrial route to prepare H-acid from Naphthalene. Also prepare a dye from 11
 H-acid.
- 3(b) Suppose you have given some naphthalene and benzene for preparing dye 09
 intermediates. Then how will you prepare the following intermediates-
 (i) Naphthalene-1,6- disulphonic acid (ii) Aniline (iii) Phthalic anhydride
- 3(c) Write down the preparation of phenol through Cumene process. 10
- 3(d) Differentiate between acid and direct dyes. 05
- 4(a) Suppose you have given some benzene. Then how will you prepare aniline yellow dye? 10
- 4(b) "Metal complex dyes improved the wash-fastness property and reduced the chance of 10
 dyestuff bleeding from leather". Explain the statement with binding mechanism of
 metal complex dye with leather fibre.
- 4(c) How will you conduct dyeing of vegetable tanned leather? 10
- 4(d) State the impact of charge on dye penetration and fixation. 05

SECTION-B

- 5(a) Suppose you have been provided with a "D" grade leather (crust) which you have to 15
 provide an opaque finish. Now formulate a standard recipe of this finishing.

5(b)	"Contact angle is essential in finish adhesion" describe the statement.	06
5(c)	Briefly explain the influencing factor of gloss retention property.	08
5(d)	Explain the polymers of finishing materials are chemically attached with the leather fibres.	06
6(a)	Write down the pros and cons of shellac as film forming material.	08
6(b)	Explain how charged ions of a dye are converted into neutral pigments?	12
6(c)	"Methyl acrylic resin is preferred over shellac as a binder". - explain why?	08
6(d)	How physical drying modifies the orientation of coating materials molecule?	07
7(a)	"The wetting and flow of dying materials depends on the surface tension and gravity". Elucidate the statement.	08
7(b)	Explain the mechanism of adhesion through dipole-dipole interaction with diagram.	07
7(c)	Write the preparation of casein through skim milk.	07
7(d)	Write down the finishing materials with their roles: (i) Plasticizers (ii) Gloss giving materials (iii) Binders and (iv) Levelling agents	13
8(a)	If a leather requires excellent flexibility and cold resistivity with strong covering properties, which binder(s) will you choose and why?	09
8(b)	Why should some chemicals not be used in water-proof finishing?	06
8(c)	Write about some important upgradation techniques of a "C" grade leather.	05
8(d)	What are the problems that may occur due to incomplete finishing?	08
8(e)	Explain the role of refractive index to observe the finished coats.	07

Department of Leather Engineering
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 B. Sc. Engineering 3rd Year 1st Term Examination-2023
LE 3109
Leather Products Engineering I

Time: 3.0 Hours

Full Marks: 210

- N.B. i) Answer any **THREE** questions from each section in separate scripts.
 ii) Figures in the right margin indicate full marks.
 iii) Assume reasonable data if any is missing.

SECTION-A

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|---|----|
| 1(a) What is clicking? State the qualities are required for a perfect clicker. | 08 |
| 1(b) Calculate the leather consumption of a complete Bi-fold wallet. If 1sq-ft leather is 200 Tk, find the total cost of a complete Bi-fold wallet. | 10 |
| 1(c) State some factors that should keep in mind during pattern placement. | 07 |
| 1(d) Why trimming allowances are important for leather goods manufacturing? | 05 |
| 1(e) How can you avoid undercut and overcut for leather goods? | 05 |
| 2(a) What are the differences among first, second and third wastages of leather products? | 06 |
| 2(b) "Factors should be considered during material consumption for leather products industries"-Explain. | 08 |
| 2(c) State the function of interlining for leather garments. | 07 |
| 2(d) Define fashion. "Fashion constantly evolves with time".-Explain. | 08 |
| 2(e) Why polypropylene (PP) cutting board is used as working surface in leather products industry? | 06 |
| 3(a) What is perspective drawing? Mention the elements of perspective drawing. Draw the inner and outer view of a Bi-fold wallet. | 10 |
| 3(b) Explain the chronological history of leather products. | 10 |
| 3(c) Write down the classification of leather products according to degree of reinforcement with example. | 08 |
| 3(d) What are the differences among interlining, interfacing and underlining? | 07 |
| 4(a) Explain the following operations during fabrication of a wallet: (i) Cradle construction (ii) Asther preparation and (iii) Centre piece. | 12 |
| 4(b) Write down the norms of pattern placement for leather goods manufacturing. | 08 |
| 4(c) What is pattern development? State the sequence of pattern development. | 08 |
| 4(d) What is lining material for leather goods? Mention the required quality for lining materials. | 07 |

SECTION-B

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|------|---|----|
| 5(a) | Suppose you are a designer of ABC leather goods Ltd. You have made three different designs of a leather jacket. Now answer the following questions:
(i) How will you finalize a design for mass production?
(ii) State and draw different types of patterns required for leather jacket manufacturing.
(iii) Describe the best suitable way to grade the center front pattern of the jacket. | 24 |
| 5(b) | “Pattern making is a bridge function between design and production”. Describe the statement. | 11 |
| 6(a) | State the information required to make the patterns of leather products. | 10 |
| 6(b) | Cite the elements of design. “Elements are the building block of a design”. –Explain with example. | 08 |
| 6(c) | Explain different types of allowances required for making leather goods. | 10 |
| 6(d) | “Basic patterns serve as a blue print of a products”. Describe the statement. | 07 |
| 7(a) | Explain the relationship between elements and principles of design. | 10 |
| 7(b) | State the different garments shape and illusions that occurred by using them. | 10 |
| 7(c) | How will you use complementary and single split complementary color schemes for dress designing? | 10 |
| 7(d) | Write down the applications of texture and line in leather products design. | 05 |
| 8(a) | State several ways to create Emphasis during product designing. | 10 |
| 8(b) | Why harmony is important in product designing? | 09 |
| 8(c) | Differentiate among classic, fashion and Fad with graphical representation. | 06 |
| 8(d) | Describe different fashion adoption ways with proper example. | 10 |

Department of Leather Engineering
Khulna University of Engineering & Technology
 B. Sc. Engineering 3rd Year 1st Term Examination-2023
LE 3101
Leather Manufacturing Technology IV

Time: 3.0 Hours

Full Marks: 210

N.B. i) Answer any **THREE** questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION-A

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| 1(a) Discuss the physical properties of leather with its control. | 12 |
| 1(b) What is meant by iso-electric point? Explain the role of iso-electric point in pickling operation. | 06 |
| 1(c) "Sheep skin is preferable for glove leather preparation"- elucidate the statement. | 08 |
| 1(d) Explain the energy and heat transfer during drying. | 09 |
| 2(a) State the properties of Sulphur-oil-vegetable tanned pickling band leather. | 09 |
| 2(b) Formulate the preparation of semi-chrome suede leather. | 06 |
| 2(c) Explain five troubleshooting cases of drying defects. | 10 |
| 2(d) Formulate and explain the use of chemical of the liming and pickling operations for full chrome suede leather. | 10 |
| 3(a) Explain the psychometric chart of drying rate. | 12 |
| 3(b) Explain the properties and modifications needed in the formulation of a football leather. | 10 |
| 3(c) How leather drying impacts on the quality of a leather? | 04 |
| 3(d) Formulate the neutralization, dyeing, and fatliquoring operation of glaze kid leather. | 09 |
| 4(a) How does moisture content vary in Chrome and Vegetable-tanned leather in relation to air humidity? | 06 |
| 4(b) Why are top dyeing and top fatliquoring done? Formulate a recipe for top dyeing and top fatliquoring for the production of garments leather from wet-blue goat skin. | 09 |
| 4(c) Define the following terms:
(i) Shoe upper leather (ii) Garments leather and (iii) Gloving leather | 06 |
| 4(d) Define comfort properties of leather. What type of information you need prior to formulating the recipe for the production of Shoe upper crust leather from wet-blue and explain why? | 14 |

SECTION-B

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|--|----|
| 5(a) Define finishing and state its role. | 07 |
| 5(b) Differentiate between ironing and plating | 04 |
| 5(c) State the parameters that are required for efficient Auto spraying. | 05 |
| 5(d) How do you improve the adhesion of the finish to leather? | 05 |

5(e)	Explain the mechanism of drying system embedded in automatic spray coating assemblies.	06
5(f)	Different finishing coats play a different role—Explain.	08
6(a)	Define the following terms: (i) Aniline finish (ii) Corrected grain finish (iii) Coating ^{and} (iv) Break ,	06
6(b)	Differentiate between laminating and plate releasing finishing.	04
6(c)	List out the prerequisites of the curst leather for finishing operation.	07
6(d)	Correlate basic chemicals used in staining and top coat with corresponding coating objectives.	08
6(e)	Illustrate the major differences among HVLP, airmix and airless spray system in a tabular form.	10
7(a)	Troubleshoot three common leather finishing defects.	08
7(b)	Write down the precautionary instructions for glazing and embossing operations.	08
7(c)	State the objectives of the following finishing operations: (i) Polishing (ii) Staking and (iii) Plating	09
7(d)	Let's say you are tasked with formulating a leather finishing recipe that provides a natural feel and excellent flexibility without covering or grain loading. What factors will you consider in designing this formulation?	10
8(a)	Write down short notes on : (i) Covering power (ii) Easy-care finish and (iii) Craquel Finish	12
8(b)	State the pros and cons of solvent based finishing	06
8(c)	State the basic working principle of a Hydraulic machine. What are the additional effects of the buffing operation?	10
8(d)	State the scope of modern milling drum.	07

Department of Leather Engineering
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 B. Sc. Engineering 3rd Year 1st Term Examination-2023
LE 3115
Analytical Chemistry for Leather Manufacture

Time: 3.0 Hours

Full Marks: 210

N.B. i) Answer any **THREE** questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION-A

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|------|---|----|
| 1(a) | Define analytical chemistry. How does analytical chemistry differ from other branches of chemistry? | 06 |
| 1(b) | Categorize the analytical chemistry in a tabular form. | 05 |
| 1(c) | Define calibration curve. The responses for blank sample of an analytical device are 0.0035, 0.0079, 0.0063, and 0.0023. Compute the limit of detection (LOD) and limit of quantification (LOQ) for that analytical device. Assume concentration unit is µg/L. | 09 |
| 1(d) | Write down the significant features of the following analytical methods: (i) Gravimetric analysis (ii) Electroanalytical (iii) Spectroscopy and (iv) Titrimetry. | 10 |
| 1(e) | With the help of a diagram describe the phase transition. | 05 |
| 2(a) | Define determinate and indeterminate error. | 04 |
| 2(b) | Describe a method in detail for chromium determination of your spent chrome liquor other than instrumental method. | 08 |
| 2(c) | Define thermogram. Briefly discuss the instrumentation of differential thermal analysis (DTA) with neat sketches. | 13 |
| 2(d) | Draw and describe a differential thermogram for sulfur where the temperature was increased ambient to above 500°C. The system was made inert by purging nitrogen gas. | 07 |
| 2(e) | Distinguish the basic difference among thermogravimetry, differential thermal analysis and differential scanning calorimetry. | 03 |
| 3(a) | Define the terms (i) Chromatography (ii) Ion Chromatography and (iii) Retention time. | 06 |
| 3(b) | Enlighten the ion separation mechanism in ion chromatography with neat sketches. | 10 |
| 3(c) | Neat sketch the ion chromatography. Briefly discuss the Mobile phase, eluent pump and stationary phase. | 15 |
| 3(d) | The calibration curve for standard sulfate (SO_4^{2-}) ion (mg/L) by ion chromatography was obtained at $y=0.0453x-0.0039$; if your tannery wastewater shows the response at the same retention time 4.0123 unit. Calculate the sulfate ion of that tannery wastewater. | 04 |
| 4(a) | Neat sketches the gas chromatography (GC). Deliberate the following terms in case of GC: i) Injection device ii) Column and iii) Detector. | 15 |
| 4(b) | Discuss the selection criteria for carrier gas in terms of GC. | 08 |

- 4(c) Differentiate between gas chromatography (GC) and high performance liquid chromatography (HPLC). 05
- 4(d) Define packed column and capillary column. Discuss the flame ionization detector (FID) with figure. 07

SECTION-B

- 5(a) Define the following chromatography terms: (i) Elute (ii) Analyte (iii) Eluent and (iv) Chromatogram. 10
- 5(b) State the basic principles of HPLC with schematic illustration. 09
- 5(c) Distinguish between isocratic and gradient elution condition in HPLC. 07
- 5(d) Write down the principles of mass spectroscopy with figures. 09
- 6(a) Give the relationship between the different components of Electromagnetic radiation (EMR). 07
- 6(b) Write on the adsorption and intensity shifts of the chemical compounds in UV-Vis spectroscopic study. 10
- 6(c) Deduce the mathematical term for Beer-Lambert law. 08
- 6(d) Write notes on barrier cell and photomultiplier tube detector used in UV-Vis spectrophotometer. 10
- 7(a) Write down the concept of molecular vibrations used in IR spectroscopy. 07
- 7(b) Write on the absorption band and wave number in IR spectroscopy. 10
- 7(c) Draw the instrumentation of IR spectrophotometer. Write the working principle of optical chopper. 08
- 7(d) Write down the construction and working principle of Michelson Interferometer used in FTIR spectrophotometer. 10
- 8(a) Distinguish between Hollow Cathode Lamp (HCL) and Electrode less Discharge Lamp (EDL) used in Atomic Absorption Spectroscopy (AAS). 08
- 8(b) Write down the premix chamber burner (PCB) atomizer in AAS with advantages and limitations. 10
- 8(c) Discuss the peak number and peak splitting aspects of NMR spectroscopy with examples. 10
- 8(d) Write notes on the solvent used to prepare sample for NMR spectroscopic analysis. 07

Department of Leather Engineering
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 B. Sc. Engineering 3rd Year 1st Term Examination-2023
LE 3111
Footwear Engineering I

Time: 3.0 Hours

Full Marks: 210

- N.B. i) Answer any **THREE** questions from each section in separate scripts.
 ii) Figures in the right margin indicate full marks.
 iii) Assume reasonable data if any missing.

SECTION-A

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|------|--|----|
| 1(a) | State the prime functions of human foot. Why human foot is unique among all living creatures? | 08 |
| 1(b) | Explain the significance of tarsus, metatarsus and phalanges in foot structure. How do these bones contribute to the overall functionality of the foot? | 10 |
| 1(c) | Identify and describe the different types of arches in the human foot. How do these arches contribute to ability of foot to support body weight and absorb shock? | 09 |
| 1(d) | Make a comparison among muscle, tendon and ligament. Why Achilles tendon is considered vital for human locomotion? | 08 |
| | | |
| 2(a) | What is meant by bunionette and how does it differ from a bunion? Discuss its management approaches relating to footwear. | 08 |
| 2(b) | Outline the motions of human foot. How do these motions integrate with different planes? | 10 |
| 2(c) | State the importance of measuring human foot for shoe making. | 07 |
| 2(d) | How can you measure different types of girth of human foot? Explain with figure. | 10 |
| | | |
| 3(a) | Discuss the procedure for using a Brannock device to measure foot. How does this device ensure a precise shoe fitting? | 10 |
| 3(b) | What is meant by shoe size? Make a list of shoe sizing system, that is based on last measurements. | 05 |
| 3(c) | Solve the following problems:
(i) Convert ladies Paris Point shoe size 39 to UK shoe size.
(ii) Convert USA ladies shoe 80 into French shoe size.
(iii) Convert Paris Point shoe size 42 to Japanese shoe size. | 09 |
| 3(d) | Describe the Paris Point shoe sizing and fitting systems with necessary tables. | 11 |
| | | |
| 4(a) | Make a comparison between the features of moccasin and slip on shoes. How does the construction of moccasins influence their cost and consumer appeal? | 06 |
| 4(b) | Sketch a basic Derby shoe with identification of its different parts. Also, write down the basic properties of a Derby shoe. | 10 |
| 4(c) | Describe the key features of sport shoes that differentiate them from other types of footwear? | 04 |
| 4(d) | Write short notes on (i) Plantar fascia (ii) Ritz stick (iii) Mondo point sizing system. | 15 |

SECTION-B

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|------|---|----|
| 5(a) | Describe briefly about the different types of rubber based soling materials that are extensively used in footwear industry. | 10 |
| 5(b) | Explain about different shoe fastening systems. | 08 |
| 5(c) | Describe the role of toe-puff and stiffener in a shoe. | 07 |
| 5(d) | Write down the complete sequence of operation for making a shoe. | 10 |
| 6(a) | Mention the ideal qualities for an insole board. Why cellulose board is widely used as insole material? | 07 |
| 6(b) | Define toe-puff. Discuss about different types of toe-puff materials. | 10 |
| 6(c) | What are the basic properties require for the shoe soling materials? | 10 |
| 6(d) | State the objectives of using reinforcement in a shoe? Mention the areas of a shoe that are needed to reinforce. | 08 |
| 7(a) | Write down the advantages and disadvantages of using crepe rubber as shoe soling material. | 10 |
| 7(b) | Write down the properties and specific application of polychloroprene adhesive. | 07 |
| 7(c) | Write down the advantage and disadvantages of using hot melt adhesive. | 10 |
| 7(d) | What are the required criteria for a sewing thread in the sewing section? | 08 |
| 8(a) | Define clicking. What qualities of a clicker are required in the cutting section of a shoe industry? | 10 |
| 8(b) | Write short no the following shoe making tools: (i) Revolving punch (ii) Lasting pincers and (iii) Cutting nippers | 07 |
| 8(c) | Define skiving. Write down the health and safety measures for skiving operation in a shoe making operation. | 10 |
| 8(d) | Write short on: (i) Open raw edge skiving (ii) Clcded raw edge skiving (iii) Shallow skiving (iv) Underlay skiving. | 08 |